



PIZZA HUT

Data Analysis

Using

SQL



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DATASET

THE ANALYSIS WAS PERFORMED ON A
RELATIONAL DATABASE FEATURING
48,000+ ROWS OF SALES DATA.

This project utilizes a comprehensive sales dataset from a pizza franchise, consisting of several interconnected tables: Orders, Order Details, Pizzas, and Pizza Types. The data encompasses thousands of transactions over a year long period, capturing critical details. By merging these relational tables, I performed deep dive analysis to uncover patterns in revenue, customer preferences, and operational efficiency.

TABLES

01

ORDERS

Order ID
Order Date
Order Time

02

ORDER DETAILS

Order Details ID
Order ID
Pizza ID
Quantity

03

PIZZAS

Pizza ID
Pizza Type ID
Size
Price

04

PIZZA TYPES

Pizza Type ID
Name
Category
Ingredients



STATEMENT

Calculate the total revenue generated from pizza sales.

OUTPUT

```
SELECT
```

```
    ROUND(SUM(order_details.quantity * pizzas.price),  
          2) AS Total_Sales
```

```
FROM
```

```
    order_details
```

```
    JOIN
```

```
    pizzas ON pizzas.pizza_id = order_details.pizza_id
```

Result Grid

	Total_Sales
▶	817860.05



STATEMENT

Identify the highest-priced pizza.

```
• SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1
```

OUTPUT

Result Grid			Filter Rows:	
	name	price		
▶	The Greek Pizza	35.95		



STATEMENT

Join the necessary tables to find the total quantity of each pizza category ordered

```
SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC
```

OUTPUT

Result Grid   Filter Rows:		
	category	quantity
▶	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050



STATEMENT

Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT
    AVG(quantity)
FROM
    (SELECT
        orders.order_date, SUM(order_details.quantity) AS quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

OUTPUT

Result Grid		
	AVG(quantity)	
▶	138.4749	



STATEMENT

Determine the top 3 most ordered pizza types based on revenue.

```
select pizza_types.name,  
sum(order_details.quantity*pizzas.price) as revenue  
from pizza_types join pizzas  
on pizza_types.pizza_type_id=pizzas.pizza_type_id  
join order_details  
on order_details.pizza_id=pizzas.pizza_id  
group by pizza_types.name  
order by revenue  
desc limit 3
```

OUTPUT

Result Grid Filter Rows:		
	name	revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5



STATEMENT

Calculate the percentage contribution of each pizza type to total revenue.

```
select pizza_types.category,  
round(sum(order_details.quantity*pizzas.price)/(select  
round(sum(order_details.quantity * pizzas.price),2) as Total_Sales  
from order_details join pizzas  
on pizzas.pizza_id = order_details.pizza_id)*100,2) as revenue  
from pizza_types join pizzas  
on pizza_types.pizza_type_id=pizzas.pizza_type_id  
join order_details  
on order_details.pizza_id=pizzas.pizza_id  
group by pizza_types.category  
order by revenue desc;
```

OUTPUT

Result Grid			Filter Rows:
	category	revenue	
▶	Classic	26.91	
	Supreme	25.46	
	Chicken	23.96	
	Veggie	23.68	



STATEMENT

Analyze the cumulative revenue generated over time.

```
select order_date,  
sum(revenue) over(order by order_date) as cum_revenue  
from  
(select orders.order_date,  
sum(order_details.quantity*pizzas.price) as revenue  
from order_details join pizzas  
on order_details.pizza_id=pizzas.pizza_id  
join orders  
on orders.order_id=order_details.order_id  
group by orders.order_date) as sales;
```

OUTPUT

Result Grid Filter Rows:			
	order_date	cum_revenue	2
▶	2015-01-01	2713.8500000000004	2
	2015-01-02	5445.75	2
	2015-01-03	8108.15	2
	2015-01-04	9863.6	2
	2015-01-05	11929.55	2
	2015-01-06	14358.5	2
	2015-01-07	16560.7	2
	2015-01-08	19399.05	2
	2015-01-09	21526.4	2
	2015-01-10	23990.350000000002	2
	2015-01-11	25862.65	2
	2015-01-12	27781.7	2
	2015-01-13	29831.300000000003	2
	2015-01-14	32358.700000000004	2
	2015-01-15	34343.500000000001	2
	2015-01-16	36937.650000000001	2
	2015-01-17	39001.750000000001	2
	2015-01-18	40978.600000000006	2